

# Physical Units Expression Language

PUEL Specification for SciNumTools v3

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# 1 PUEL Specification

## 1.1 Overview

PUEL (Physical Units Expression Language) is a domain-specific language for representing and manipulating physical units and dimensional expressions.

It defines the syntax and structure of unit expressions used in scientific computations within the SciNumTools3 framework. Evaluation, normalization, dimensional analysis, and unit conversion are performed by the underlying EXS and PUQ modules.

The language is designed for the following use cases:

- As a component of DIPL for defining and validating unit-aware parameters.
- As a standalone language for the concise and machine-readable representation of physical unit expressions.
- As a backend for programmatic interfaces, such as Python bindings and other language integrations.

This chapter defines the language syntax and conversion rules. The following chapters provide reference tables for scaling prefixes, unit systems, and shared definitions, followed by the derived units, constants, and quantity scales of each supported system.

Focused normative sections define validity and errors, uncertainty propagation, and special-unit semantics and conformance options.

### 1.1.1 Terminology

A **unit expression** describes units or dimensions, such as `kg*m*s-2`. A **quantity expression** additionally supplies a value, such as `12*kg*m*s-2`. Both are standalone, whitespace-free PUEL expressions. A **calculator expression**, such as `12*kg + 3*g`, is a PUQ operation over complete PUEL quantities and is not part of the standalone PUEL grammar. Host APIs may also construct quantities from numerical arrays or values without first parsing a PUEL literal.

---

## 1.2 Design Principles

PUEL is designed to be:

- **Concise** — compact textual representation of units
- **Composable** — units can be combined using algebraic operations
- **Deterministic** — expressions resolve to an unambiguous dimensional meaning
- **Extensible** — supports custom units and prefixes
- **Solver-backed** — evaluation delegated to EXS

### 1.2.1 Relationship to Existing Unit Standards

PUEL is a machine-readable language for representing physical quantities and unit expressions, rather than a replacement for established unit coding systems. Its vocabulary and dimensional model are designed to be understandable alongside SI, UCUM, UDUNITS, QUDT, and the OM ontology. PUEL additionally expresses numerical values, uncertainties, physical constants, symbolic quantities, and algebraic unit factors.

SciNumTools3 does not currently claim complete bidirectional conversion to those external serializations. A PUEL expression remains semantically independent of any particular external notation or ontology.

### 1.2.2 Portable PUEL profile

For exchange between implementations, producers SHOULD use the portable profile: ASCII syntax, SI linear units, no custom symbols, no system-specific constants or quantity scales, and no temperature or logarithmic units. Use compact array literals without spaces, such as `[2,3,4]*m`, when arrays are needed. Producers using an extended feature SHOULD record the required unit system and feature family with the exchanged data.

---

## 1.3 Unit Model

### 1.3.1 Base units

All unit expressions are ultimately represented in terms of a fixed set of base dimensions:

- Length in meters (**m**)
- Mass in grams (**g**)
- Time in seconds (**s**)
- Temperature in kelvins (**K**)
- Electric current in amperes (**A**)

- Luminous intensity in candelas (**cd**)
- Amount of substance in moles (**mol**)
- Angle in radians (**rad**)

Derived units are represented as combinations of these base dimensions together with an associated scaling factor.

The choice of base dimensions is driven by implementation considerations and does not exactly match the SI base-unit system.

- Mass is represented using grams (**g**) rather than kilograms (**kg**) to avoid embedding a prefix in a base unit.
- Electric current is represented using amperes (**A**); electric charge is a derived quantity (**C = A\*s**).
- Angle is included as a base dimension to enable dimensional validation of mathematical functions that require angular arguments. However, angular units **MUST** remain convertible to dimensionless units, reflecting their dimensionless nature in the SI system.

### 1.3.2 Scaling factors

Scaling prefixes provide a compact way to represent decimal multiples and submultiples of units. A scaling prefix modifies a unit identifier by applying a fixed numerical factor while preserving the unit’s dimensionality.

For example, the expressions **m**, **cm**, and **km** all represent units of length and differ only in their scaling factors relative to the base unit.

PUEL defines a standard set of scaling prefixes. Implementations **MUST** recognize and correctly apply these prefixes when parsing unit expressions. However, prefixes **SHOULD** only be permitted for units for which their use is conventional or widely accepted.

The complete list of supported scaling prefixes is provided in Scaling Prefixes.

### 1.3.3 Named Entities

Named entities in PUEL include all units (e.g., Joule **J**, degree Celsius **Cel**, decibel **dB**), physical constants (e.g., fixed proton mass **{#m\_p}**, system-specific Avogadro constant **{N\_A}**, solar luminosity **{L\_sol}**), physical quantities (e.g., energy **<E>**, electric flux **<Phi\_E>**, radiation dose **<D\_r>**), and unit-system scaling factors (e.g., **|E|**, **|Phi\_E|**, **|D\_r|**). The dimensionality of these entities is defined in terms of the base dimensions described in the previous section.

PUEL distinguishes between these categories using the following notation:

- Units are represented by their symbols: **J**, **Cel**, **dB**
- Physical constants are enclosed in curly braces. A leading **#** denotes a fixed reference constant, such as **{#m\_p}**; without it, the constant is resolved in the active unit system, such as **{N\_A}** or **{L\_sol}**.
- Physical quantities are enclosed in angle brackets: **<E>**, **<Phi\_E>**, **<D\_r>**
- Unit-system scaling factors are enclosed in pipe symbols: **|E|**, **|Phi\_E|**, **|D\_r|**

Unit-system scaling factors represent the numerical coefficients required to convert a physical quantity between the SI system and another system of units. They are primarily intended to support conversions between systems with different dimensional conventions, such as SI and CGS.

Each implementation **MUST** define the numerical values and dimensional representations of all supported derived entities. These definitions **SHOULD** be precomputed and stored to avoid repeated evaluation and dimensional derivation during the processing of unit expressions.

Within a given implementation and unit system, all supported identifiers and scaling-prefix combinations **MUST** be unique. Consequently, every valid token in a PUEL expression **MUST** have exactly one interpretation.

### 1.3.4 Custom units

Custom units **MAY** be defined in the surrounding environment using existing unit expressions. Once registered in a unit system, a custom symbol can be used in expressions just like a built-in unit, including in products, quotients, and powers. For example, an environment may define **step** as **0.75\*m** and then interpret **4\*step** as **3\*m**.

Custom definitions are scoped to their unit system. PUQ stores their numerical scale, uncertainty when present, and dimensions. Registration is performed by the host application; PUEL does not define assignment syntax. PUQ custom units do not accept scaling prefixes.

---

## 1.4 Unit expressions

### 1.4.1 General Form

A unit expression consists of an optional unit system identifier followed by a numerical value, unit factors, or both.

General form:

[<system>\_] [<value>\*] [<unit-expression>]

where:

- <system> is a unit system identifier.
- <value> is a numerical value, optionally including an uncertainty.
- <unit-expression> is a combination of unit identifiers and algebraic operators.

At least one of <value> or <unit-expression> MUST be present.

The \* separator between the value and unit factors is present only when both are supplied. A value alone, such as 2.5, represents a dimensionless quantity. The value MAY also be a numerical array, as described below.

Examples:

```
m
2.5*m
3.452(3)*kg*m/s2
US_lb*ft
SI_9.81*m/s2
```

The following rules apply in unit expressions:

| Rule                                                                                                                                                                                                                         | Examples                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Expressions MUST NOT contain whitespace. PUQ MAY accept one space after an array comma as an input convenience, but portable and canonical array literals omit it.                                                           | kg*m/s2; [2,3,4]*m                                                                                                    |
| Unit identifiers MAY be combined using multiplication (*), division (/), and grouping parentheses ((), ).                                                                                                                    | kg*m; m/s; kg*m2/(sr*s2)                                                                                              |
| Exponents are written directly after the corresponding unit identifier and do not use exponentiation operators such as ^ or **.                                                                                              | m2                                                                                                                    |
| Fractional exponents are written using the <b>numerator:denominator</b> notation.                                                                                                                                            | kg3:2                                                                                                                 |
| Negative exponents are written using a leading minus sign.                                                                                                                                                                   | s-1                                                                                                                   |
| A unit system MAY be specified. When present, it MUST appear at the beginning of the expression and MUST be separated from the remainder by an underscore (_).                                                               | US_lb*ft                                                                                                              |
| Numerical values MAY be included in expressions.                                                                                                                                                                             | 2.34e3*mol; -9.81*m/s2                                                                                                |
| The symbols + and - denote the sign of a numerical value or exponent; they do not represent addition or subtraction operators.                                                                                               | +5*m; -5*m; s-2                                                                                                       |
| Uncertainties are written in parentheses immediately following the last decimal digit of a numerical value. The number of digits in the uncertainty corresponds to the same number of least significant digits in the value. | 3.45234(2)e3 → 3452.34 ± 0.02;<br>3.45234(12) → 3.45234 ± 0.00012;<br>12.3(4)e-2 → 0.123 ± 0.004; 120(5)<br>→ 120 ± 5 |

### 1.4.2 Expression Parsing

Although numerical values are typically written at the beginning of an expression, they are not restricted to that position. Numerical factors MAY appear anywhere within a unit expression as multiplicative constants. For example, 2\*kg\*m/s2 and kg\*(2\*m)/s2 are equivalent.

Implementations SHOULD accept numerical factors in any valid position within a unit expression. However, producers of unit expressions SHOULD place any numerical factor at the beginning of the expression, as this improves readability and simplifies parsing.

Unit expressions MAY use division operators and grouping parentheses for readability. For example, `kg*m2/(sr*s2)` and `kg*m2*sr-1*s-2` represent the same unit expression.

An exponent MAY follow a closing parenthesis and applies to the entire grouped expression, including numerical factors. Integer, negative, and fractional exponents use the same notation as unit exponents. For example, `(m/s)2` is equivalent to `m*s-2`, and `kg/(m*s)-1:2` is equivalent to `kg*m1:2*s1:2`. Exponents also apply to named entities, as in `kg*<v>2`.

Expressions containing only numerical factors are supported. For example, `3*(2.0e1/5.0)` evaluates to the dimensionless value 12.

#### 1.4.3 Numerical arrays

A numerical array is written as a comma-separated list of values in square brackets. All elements share the units specified outside the brackets. Elements MAY include scientific notation or parenthesized uncertainties. The portable form contains no spaces; PUQ also accepts one optional space after a comma for input compatibility.

```
[2,3.4,5e6]*km/s
[2.00(20),3.00(30)]*m
```

Multiplication and division by a scalar apply to each element. Multiplication and division between arrays of the same shape are elementwise operations. For example, `[20,40.5]*2` evaluates to `[40,81]`, and `[20,40.5]*[2,3]` evaluates to `[40,121.5]`. Unit factors are combined using the ordinary expression rules.

#### 1.4.4 Unit-system selection

The system identifier, when present, MUST appear only at the beginning of the expression, immediately followed by an underscore (`_`). It selects the interpretation of all unit factors in the expression. For example, `US_23*ft2` is valid; `US 23*ft2` and `23*US_ft2` are not valid PUEL expressions. PUQ supports SI and, when enabled, CGS variants (ESU, EMU, GU), English systems (IU, US), and natural systems (AU, PU, SRU, GRU, GEO).

When no identifier is supplied, PUQ uses the active unit system of the surrounding environment, initially SI. If both the expression and its environment explicitly select a system for that expression, the selections MUST agree. Producers SHOULD include the system identifier when an expression must retain its interpretation outside its original environment.

#### 1.4.5 Display notation

In addition to ASCII notation, PUQ can produce Unicode and HTML representations for display. When Unicode input preprocessing is enabled, it accepts superscript exponents, the Unicode middle dot (U+22C5) for multiplication, the Unicode minus sign (U+2212), and the multiplication sign (U+00D7) for scientific notation. For example, a displayed form using U+00D7 before 10 and superscript U+2212/U+22C5 symbols corresponds to the portable expression `6.23537e-2*C3`.

Unicode input conveniences are optional; ASCII notation remains the portable form. A system identifier MUST retain its leading position and underscore separator even when Unicode notation is used, as in the displayed form of `US_23*ft2`. HTML output is a display format, not an expression input format.

#### 1.4.6 Normalization and Serialization

Implementations MUST resolve a valid expression to its numerical scale, dimensions, and applicable special-unit semantics. Division may be normalized to negative exponents; for example, `kg*m2/(sr*s2)` and `kg*m2*sr-1*s-2` have the same dimensional meaning.

PUEL does not yet prescribe one unique textual serialization for every semantically equivalent expression. Implementations may retain factor order or choose their own normalized display form. They MUST NOT imply that formatting alone preserves the original placement of division operators or grouping parentheses.

For portable serialization, producers SHOULD use ASCII notation, place a numerical factor first, join factors with `*`, use negative exponents rather than `/`, and omit whitespace from array literals. Implementations that offer a stronger canonical serialization MUST document their factor ordering, numeric precision, uncertainty formatting, prefix rebasing, and treatment of temperature and logarithmic units.

When representing base dimensions, only dimensions with non-zero exponents SHOULD be included. Their order MUST follow the base-unit list in Base units: `m`, `g`, `s`, `K`, `A`, `cd`, `mol`, and `rad`.

## 1.5 Calculator expressions

The grammar in `grammar.ebnf` defines a single, whitespace-free PUEL unit expression. `Quantity` and `UnitSolver` parse that form directly; for example, write `12*km`, not `12 km`.

PUQ also provides a calculator layer for arithmetic between complete PUEL quantities. Its binary operators are written with spaces around them, so each operand remains a valid PUEL expression:

```
23*cm + 3*m
2*g * ( 6*m2/s2 - 3*cm2 / 1.5*s2 )
```

The calculator supports grouping, unary `+` and `-`, and binary `+`, `-`, `*`, and `/`, in the usual precedence order. Calculator syntax is a PUQ host operation rather than an additional form accepted by the standalone PUEL unit solver. Likewise, the C++ `puq::math` functions such as `sqrt`, `pow`, `log`, and trigonometric functions operate on parsed quantities but do not add function-call syntax to PUEL.

## 1.6 Conversions

PUQ conversion APIs operate on quantities parsed from PUEL and can convert between units of identical dimensions:

```
km → m
eV → J
dBW → W
```

Special handling exists for:

- logarithmic units
- temperature units

These require custom conversion rules.

### 1.6.1 Logarithmic and temperature conversions

Logarithmic conversions distinguish ratios from levels with a defined reference. PUQ supports power ratios (PR), amplitude ratios (AR), and logarithmic units such as decibels (dB) and nepers (Np). Power levels such as dBm and dBW include their reference power: `0*dBm` corresponds to `1*mW`, while `0*dBW` corresponds to `1*W`. A bare dB value does not specify an absolute power.

Absolute temperature conversions can include an offset as well as a scale. For example, `23*Cel` converts to `296.15*K`. PUQ supports Celsius (`Cel`), Fahrenheit (`degF`), kelvin (K), and Rankine (`degR`). Such conversions cannot be described by a multiplicative scaling factor alone.

### 1.6.2 Conversion between unit systems

Conversions between systems with different dimensional conventions may require a physical-quantity context in addition to source and target units. For example, converting SI current in A to electrostatic CGS current in `statA` requires the current context I. Energy conversion between J and `erg` does not require an explicit context.

The context identifies the corresponding physical quantity in each system and enables the use of its unit-system scaling factors. It is supplied by the surrounding application, not appended as an additional operator in the expression. A system prefix selects a system; it does not by itself specify a conversion context.

## 2 Validity and Errors

This section defines when a PUEL expression is valid independently of the particular API used to parse it.

### 2.1 Lexical and structural validity

A standalone PUEL expression MUST conform to the grammar in `grammar.ebnf`. In particular:

- it MUST contain at least one value, unit factor, named entity, or grouped expression;
- whitespace is forbidden, including after an array comma;
- an array literal is one-dimensional and MUST contain at least one scalar value;
- a fractional exponent denominator MUST be non-zero;
- a system identifier, when present, MUST be the first token and MUST be followed by `_`; and

- + and - are signs only in standalone PUEL. Addition and subtraction belong to the spaced PUQ calculator syntax.

An implementation **MUST** reject malformed delimiters, unknown identifiers, invalid numerical literals, malformed uncertainties, and unbalanced grouping parentheses. It **MUST NOT** silently reinterpret an unknown identifier as a prefix-plus-unit combination.

## 2.2 Vocabulary and system validity

Each unit, prefix combination, constant, quantity scale, and system factor **MUST** resolve unambiguously in the selected unit system. A prefix is valid only when the selected unit explicitly permits that prefix. Custom symbols follow the same uniqueness rule and do not accept prefixes in the PUQ reference implementation. PUQ may accept an optional space after a literal array comma as a non-portable input convenience.

Without an explicit system identifier, resolution uses the active unit system. If an API separately supplies a system and the expression contains a different system identifier, the expression is invalid because its interpretation would be ambiguous.

## 2.3 Dimensional and shape validity

Parsing establishes dimensions but does not by itself request a conversion. Operations and conversions impose the following additional requirements:

- addition and subtraction require compatible dimensions;
- conversion requires compatible dimensions, unless a documented cross-system quantity context is supplied;
- a quantity used as an exponent in a PUQ mathematical operation **MUST** be dimensionless; and
- elementwise array operations require compatible array shapes.

The literal PUEL grammar deliberately defines only one-dimensional arrays. PUQ can also process multidimensional values supplied through the host VAL/C++ API; that capability is not a multidimensional PUEL literal syntax.

## 2.4 Diagnostics

Implementations **SHOULD** report the failing expression or unit, the reason it is invalid, and a corrective action. The PUQ reference implementation distinguishes parse failures, unknown or disallowed prefixes, unit-system conflicts, dimension mismatch, and unavailable special-unit conversions.

# 3 Uncertainty Propagation

A scalar PUEL uncertainty is written immediately after its estimate, for example `3.452(3)*kg`. The parenthesized digits represent an **absolute** uncertainty in the last significant digits of the estimate. Thus `3.452(3)` represents  $3.452 \pm 0.003$ , while `3.452(3)e3` represents  $3452 \pm 3$ .

Array literals may attach an uncertainty to each element:

```
[2.00(20),3.00(30)]*m
```

The estimate and uncertainty arrays **MUST** have the same shape. A quantity without parenthesized uncertainty has no stored uncertainty.

## 3.1 Reference PUQ propagation model

PUQ stores absolute uncertainty in the same units as the estimate. Linear unit conversion scales both estimate and uncertainty by the conversion factor. Offset temperature conversion changes the estimate by its offset while scaling the uncertainty only by the scale.

For addition and subtraction, the reference implementation uses a conservative sum of absolute uncertainties:

$$(x \pm dx) \pm (y \pm dy) = (x \pm y) \pm (dx + dy)$$

For multiplication and division, it evaluates the extrema formed by the input intervals and retains the larger absolute deviation from the central result. This is an interval-style bound, not Gaussian root-sum-square propagation.

The public PUQ mathematical functions use first-order derivative propagation where applicable. For example, for a constant exponent  $p$ :

$$y = x^p$$

$$dy = |p * x^{(p - 1)}| * dx$$

For a variable exponent, the independent contributions are combined by root-sum-square. Logarithmic conversion and mathematical functions propagate uncertainty through their corresponding transformation.

PUQ does not model correlations between input uncertainties. Applications requiring covariance-aware or statistical uncertainty analysis MUST provide that model outside the PUEL expression language.

## 4 Special Units and Conformance Options

Most PUEL units are linear: they convert through a scale factor and can be freely combined by unit algebra. Temperature and logarithmic units are different. They are valid PUEL symbols, but conversion requires specialized semantics.

### 4.1 Temperature units

K, Cel, degF, and degR represent absolute temperature scales. A conversion may contain both a scale and an offset:

$$23 * \text{Cel} = 296.15 * \text{K}$$

Consequently, Cel is not a multiplicative synonym for K. When a temperature factor occurs with one ordinary linear factor, PUQ can apply its specialized conversion to expressions such as Cel/m. Rebase-to-base-dimensions is not defined for expressions containing temperature units.

### 4.2 Logarithmic units

Logarithmic ratios and levels include AR, PR, B, dB, Np, dBm, dBW, and the reference levels listed in the unit-system tables. A ratio has no absolute reference; a level does. For example:

$$0 * \text{dBm} = 1 * \text{mW}$$

$$0 * \text{dBW} = 1 * \text{W}$$

The complete tables define each level's reference quantity. PUQ supports a single logarithmic factor with an optional ordinary linear factor, for example dBmW/Hz. It does not define arbitrary algebraic rebasing of logarithmic units; rebase-to-base-dimensions is invalid for expressions containing one.

### 4.3 Conversion availability

Matching physical dimensions are necessary but not sufficient for a special conversion. The source and target special-unit combination must have a defined conversion rule. Otherwise a converter MUST report that the conversion is unavailable rather than applying a linear scale factor.

For conversions between systems whose dimensional conventions differ, a physical-quantity context may also be required. For example, I identifies electric current when converting between statA and A; energy conversion between erg and J does not require that context.

### 4.4 Reference implementation feature matrix

The PUQ reference implementation always provides SI. Its compile-time configuration can enable or disable the following optional feature families:

| Feature family    | PUEL vocabulary affected                                   |
|-------------------|------------------------------------------------------------|
| CGS systems       | ESU, EMU, and GU units and quantity scales                 |
| English systems   | IU and US units                                            |
| Natural systems   | AU, PU, SRU, GRU, and GEO units and quantity scales        |
| Temperature units | Cel, degF, and degR with offset conversions                |
| Logarithmic units | ratios, bel/neper units, and referenced logarithmic levels |

| Feature family        | PUEL vocabulary affected                                            |
|-----------------------|---------------------------------------------------------------------|
| Unicode preprocessing | display-oriented Unicode input aliases; portable PUEL remains ASCII |

The SciNumTools3 default configuration enables these feature families. A consumer that exchanges PUEL expressions across implementations SHOULD either use the portable SI linear subset or record the required systems and special unit features with the data.

## 5 Scaling prefixes

| Prefix | Factor | Name  |
|--------|--------|-------|
| Y      | 1e24   | yotta |
| Z      | 1e21   | zetta |
| E      | 1e18   | exa   |
| P      | 1e15   | peta  |
| T      | 1e12   | tera  |
| G      | 1e9    | giga  |
| M      | 1e6    | mega  |
| k      | 1e3    | kilo  |
| h      | 1e2    | hecto |
| da     | 1e1    | deka  |
| d      | 1e-1   | deci  |
| c      | 1e-2   | centi |
| m      | 1e-3   | milli |
| u      | 1e-6   | micro |
| n      | 1e-9   | nano  |
| p      | 1e-12  | pico  |
| f      | 1e-15  | femto |
| a      | 1e-18  | atto  |
| z      | 1e-21  | zepto |
| y      | 1e-24  | yocto |

[!NOTE] The prefix u is used to represent micro (Unicode U+03BC) in ASCII-only unit expressions.

## 6 Unit systems

The following tables specify the built-in vocabulary provided by PUQ for PUEL expressions. A system identifier appears only at the beginning of an expression and is immediately followed by \_, for example SI\_2\*J or US\_2\*gal. An identifier selects the entire expression; it cannot qualify an individual factor.

| Identifier | System                        | Unit and quantity tables |
|------------|-------------------------------|--------------------------|
| SI         | International System of Units | SI definitions           |
| ESU        | Electrostatic CGS units       | ESU definitions          |
| EMU        | Electromagnetic CGS units     | EMU definitions          |
| GU         | Gaussian CGS units            | GU definitions           |
| IU         | Imperial units                | IU definitions           |
| US         | United States customary units | US definitions           |
| AU         | Atomic units                  | AU definitions           |
| PU         | Planck units                  | PU definitions           |
| SRU        | Units for special relativity  | SRU definitions          |
| GRU        | Units for general relativity  | GRU definitions          |
| GEO        | Geometrized units             | GEO definitions          |

## 6.1 Reading the tables

- Every system includes the shared base-unit symbols. Each system page lists its additional units and aliases, constants, and quantity scales. A unit available in SI is not automatically available in another system.
- Definitions describe one unprefix unit in terms of symbols from the same system. Prefix permissions apply to the symbol in that row; prefixes are not implicitly inherited through an alias.
- Numerical factors and constant values reproduce the PUQ definitions, including their stated precision. These tables describe the PUEL vocabulary and do not redefine external metrological standards.
- Temperature and logarithmic rows give reference units, not linear equalities. Use the special conversion rules; for example, **Ce1** is not equal to K.
- SI is always available in PUQ. Other system families, offset temperature units, and logarithmic units depend on enabled implementation features.
- Natural systems often use quantity scales such as <E> rather than a new named unit. Constants without # may be normalized to unity; fixed {#. . .} reference constants retain their reference values.

## 7 Shared unit definitions

These symbols are available in every listed system. Prefixes are case-sensitive: **All** means any prefix in the prefix table, **None** means no prefix, and an explicit list permits only those prefixes. An unprefix symbol is always permitted.

### 7.1 Base-unit symbols

| Symbol | Meaning | Definition or reference | Prefixes |
|--------|---------|-------------------------|----------|
| m      | meter   | m                       | All      |
| g      | gram    | g                       | All      |
| s      | second  | s                       | All      |
| K      | Kelvin  | K                       | All      |
| A      | Ampere  | A                       | All      |
| cd     | candela | cd                      | All      |
| mol    | mole    | mol                     | All      |
| rad    | radian  | rad                     | m        |

### 7.2 Fixed reference constants

The following constants are available in SI, AU, PU, SRU, GRU, and GEO. The # inside the braces distinguishes a fixed reference constant from a system-specific constant: for example, {#c} retains its reference value while {c} may be 1 in a natural system. Parenthesized digits denote uncertainty using the ordinary PUEL notation.

| Symbol   | Meaning          | Definition or reference             | Prefixes |
|----------|------------------|-------------------------------------|----------|
| {#alpha} | fine str. const. | 7.2973525643(11)e-3                 | None     |
| {#euler} | Euler's num.     | 2.718281828459                      | None     |
| {#N_0}   | Avogadro's num.  | 6.02214076e23                       | None     |
| {#pi}    | pi num.          | 3.1415926535                        | None     |
| {#a_0}   | Bohr radius      | 5.29177210544(82)e-11*m             | None     |
| {#c}     | speed of light   | 299792458*m/s                       | None     |
| {#e}     | elem. charge     | 1.602176634e-19*A*s                 | None     |
| {#eps_0} | permit. of vac.  | 8.8541878188(14)e-12*m-3*kg-1*s4*A2 | None     |
| {#G}     | grav. const.     | 6.67430(15)e-11*m3*kg-1*s-2         | None     |
| {#g}     | grav. accel.     | 9.80665*m/s2                        | None     |
| {#h}     | Planck const.    | 6.62607015e-34*m2*kg*s-1            | None     |
| {#hbar}  | Reduced Pl. con. | 1.054571817e-34*m2*kg*s-1           | None     |
| {#k_B}   | Boltzmann const. | 1.380649e-23*m2*kg*s-2*K-1          | None     |
| {#mu_0}  | permeab. of vac. | 1.25663706127(20)e-6*m*kg*s-2*A-2   | None     |
| {#mu_B}  | Bohr magneton    | 9.2740100657(29)e-24*m2*A-1         | None     |
| {#m_e}   | electron mass    | 9.1093837139(28)e-31*kg             | None     |
| {#m_p}   | proton mass      | 1.67262192595(52)e-27*kg            | None     |

| Symbol   | Meaning           | Definition or reference                                             | Prefixes |
|----------|-------------------|---------------------------------------------------------------------|----------|
| {#m_n}   | neutron mass      | $1.67492749804(95)e-27\text{kg}$                                    | None     |
| {#R_inf} | Rydberg constant  | $10973731.568157(12)*m^{-1}$                                        | None     |
| {#sigma} | Stef.Bolt. const. | $5.670374419e-8\text{kg*s}^{-3}\text{K}^{-4}$                       | None     |
| {#N_A}   | Avogadro's const. | $6.02214076e23/\text{mol}$                                          | None     |
| {#kap_0} | permittivity      | $1.11265005620e-13\text{m}^{-3}\text{g}^{-1}\text{s}^4\text{A}^2$   | None     |
| {#k_e}   | Coulomb const.    | $8.9875517923(14)e9\text{m}^3\text{kg}^{-1}\text{s}^4\text{A}^{-2}$ | None     |
| {#H_0}   | Hubble const.     | $67.36(54)*3.24076e-20\text{s}^{-1}$                                | None     |
| {#L_sol} | Solar luminosity  | $3.828e26\text{m}^2\text{kg*s}^{-3}$                                | None     |
| {#M_sol} | Solar mass        | $1.988475(92)e30\text{kg}$                                          | None     |
| {#R_sol} | Solar radius      | $6.9566(14)e8\text{m}$                                              | None     |

## 8 SI: International System of Units

Select this system with the leading `SI_` prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

### 8.1 Derived units and aliases

| Symbol | Meaning          | Definition or reference       | Prefixes                     |
|--------|------------------|-------------------------------|------------------------------|
| au     | astr. unit       | $149597.870691\text{Mm}$      | None                         |
| AU     | astr. unit       | au                            | None                         |
| ly     | light-year       | {c}*yr_j                      | k, M, G                      |
| pc     | parsec           | $3.0857e16\text{m}$           | k, M, G, T                   |
| Å      | Angstrom         | $1e-10\text{m}$               | m, k                         |
| u      | atomic mass unit | $\text{g}/\{\text{N}_0\}$     | None                         |
| amu    | atomic mass unit | u                             | None                         |
| Da     | Dalton           | u                             | None                         |
| t      | tonne            | $1e3\text{kg}$                | k, m, G                      |
| ct     | carat            | $200\text{mg}$                | None                         |
| min    | minute           | $60\text{s}$                  | None                         |
| h      | hour             | $60\text{min}$                | None                         |
| day    | day              | $24\text{h}$                  | None                         |
| yr_t   | tropical year    | $365.24219\text{day}$         | k, m, G                      |
| yr_j   | Julian year      | $365.25\text{day}$            | k, m, G                      |
| yr_g   | Gregorian year   | $365.2425\text{day}$          | k, m, G                      |
| yr     | year             | yr_j                          | k, m, G                      |
| degR   | degree Rankine   | $5/9\text{K}$                 | None                         |
| deg    | angle degree     | $2*\{\pi\}*\text{rad}/360$    | None                         |
| '      | angle minute     | $\text{deg}/60$               | None                         |
| ''     | angle second     | $'/60$                        | None                         |
| ar     | are              | $100\text{m}^2$               | c, d, da, h                  |
| b      | barn             | $1e-28\text{m}^2$             | M, k, m, u, n, p, f, a, z, y |
| l      | liter            | $\text{dm}^3$                 | All                          |
| L      | liter            | l                             | All                          |
| J      | Joule            | $\text{N*m}$                  | All                          |
| eV     | electronvolt     | {e}*V                         | All                          |
| erg    | erg              | $\text{dyn*cm}$               | None                         |
| cal    | calorie          | $4.184\text{J}$               | k, M                         |
| Cal    | Calorie          | kcal                          | None                         |
| Ha     | Hartree          | $4.3597447222071e-18\text{J}$ | k, M                         |
| E_h    | Hartree          | Ha                            | k, M                         |
| Pa     | Pascal           | $\text{N/m}^2$                | All                          |
| atm    | atm. pressure    | $101325\text{Pa}$             | None                         |
| bar    | bar              | $100\text{kPa}$               | M, k, d, c, m                |

| Symbol   | Meaning           | Definition or reference                                | Prefixes |
|----------|-------------------|--------------------------------------------------------|----------|
| Ba       | Barye             | $0.1 \cdot \text{Pa}$                                  | None     |
| N        | Newton            | $\text{kg} \cdot \text{m} / \text{s}^2$                | All      |
| dyn      | dyne              | $\text{g} \cdot \text{cm} / \text{s}^2$                | All      |
| Gy       | Gray              | $\text{J} / \text{kg}$                                 | None     |
| Sv       | Sivert            | $\text{J} / \text{kg}$                                 | None     |
| T        | Tesla             | $\text{Wb} / \text{m}^2$                               | All      |
| G        | Gauss             | $10^{-4} \cdot \text{T}$                               | All      |
| Hz       | Hertz             | $\text{s}^{-1}$                                        | All      |
| Bq       | Becquerel         | $\text{s}^{-1}$                                        | None     |
| W        | Watt              | $\text{J} / \text{s}$                                  | All      |
| hp       | horse power       | $745.7 \cdot \text{W}$                                 | None     |
| kph      | kilometer p. hour | $\text{km} / \text{h}$                                 | None     |
| kn       | knot              | $0.514444 \cdot \text{m} / \text{s}$                   | None     |
| Mx       | Maxwell           | $10^{-8} \cdot \text{Wb}$                              | None     |
| Wb       | Weber             | $\text{V} \cdot \text{s}$                              | None     |
| P        | Poise             | $\text{g} / (\text{cm} \cdot \text{s})$                | c        |
| St       | Stokes            | $\text{cm}^2 / \text{s}$                               | c        |
| Ka       | Kayser            | $\text{cm}^{-1}$                                       | None     |
| D        | Debye             | $3.33564 \cdot 10^{-30} \cdot \text{C} \cdot \text{m}$ | All      |
| Oe       | Oersted           | $10^3 \cdot \text{A} / (4 \cdot \pi \cdot \text{m})$   | None     |
| Gal      | Gal               | $\text{cm} / \text{s}^2$                               | None     |
| Bi       | Biot              | $10 \cdot \text{A}$                                    | None     |
| Rad      | radiation dose    | $0.01 \cdot \text{Gy}$                                 | None     |
| sr       | steradian         | $\text{rad}^2$                                         | None     |
| lm       | lumen             | $\text{cd} \cdot \text{sr}$                            | None     |
| C        | coulomb           | $\text{A} \cdot \text{s}$                              | All      |
| H        | Henry             | $\text{Wb} / \text{A}$                                 | All      |
| Ohm      | Ohm               | $\text{V} / \text{A}$                                  | All      |
| V        | Volt              | $\text{J} / \text{C}$                                  | All      |
| lx       | lux               | $\text{lm} / \text{m}^2$                               | None     |
| F        | Farad             | $\text{C} / \text{V}$                                  | All      |
| S        | Siemens           | $\text{Ohm}^{-1}$                                      | All      |
| kat      | katal             | $\text{mol} / \text{s}$                                | All      |
| dpt      | dioptré           | $1 / \text{m}$                                         | All      |
| Mo       | seismic moment    | $\text{N} \cdot \text{m}$                              | All      |
| %        | percent           | $10^{-2}$                                              | None     |
| permille | promile           | $10^{-3}$                                              | None     |
| ppth     | promile           | $10^{-3}$                                              | None     |
| ppm      | parts per million | $10^{-6}$                                              | None     |
| ppb      | parts per billion | $10^{-9}$                                              | None     |
| Cel      | degree Celsius    | K (temperature reference;<br>offset conversion)        | None     |
| degF     | degree Fahrenheit | K (temperature reference;<br>offset conversion)        | None     |
| PR       | Power ratio       | 1                                                      | None     |
| AR       | Amplitude ratio   | 1                                                      | None     |
| Np       | Nepers            | 1 (logarithmic reference;<br>special conversion)       | c, d     |
| B        | Bel               | 1 (logarithmic reference;<br>special conversion)       | d        |
| Bm       | bel-milliwatt     | mW (logarithmic reference;<br>special conversion)      | d        |
| BmW      | bel-milliwatt     | mW (logarithmic reference;<br>special conversion)      | d        |
| BW       | bel-watt          | W (logarithmic reference;<br>special conversion)       | d        |
| BV       | bel-volt          | V (logarithmic reference;<br>special conversion)       | d        |

| Symbol | Meaning                     | Definition or reference                                      | Prefixes |
|--------|-----------------------------|--------------------------------------------------------------|----------|
| BuV    | bel-microvolt               | uV (logarithmic reference; special conversion)               | d        |
| BA     | bel-amps                    | A (logarithmic reference; special conversion)                | d        |
| BuA    | bel-microamps               | uA (logarithmic reference; special conversion)               | d        |
| B0hm   | bel-ohms                    | Ohm (logarithmic reference; special conversion)              | d        |
| BSPL   | bel-SPL (Pa)                | Pa (logarithmic reference; special conversion)               | d        |
| BSIL   | bel-SIL (W/m <sup>2</sup> ) | W/m <sup>2</sup> (logarithmic reference; special conversion) | d        |
| BSWL   | bel-SWL (W)                 | W (logarithmic reference; special conversion)                | d        |
| Mw     | mom. magnitude              | Mo (logarithmic reference; special conversion)               | None     |

## 8.2 System-specific constants

| Symbol  | Meaning           | Definition or reference | Prefixes |
|---------|-------------------|-------------------------|----------|
| {alpha} | fine str. const.  | {#alpha}                | None     |
| {euler} | Euler's num.      | {#euler}                | None     |
| {N_0}   | Avogadro's num.   | {#N_0}                  | None     |
| {pi}    | pi num.           | {#pi}                   | None     |
| {a_0}   | Bohr radius       | {#a_0}                  | None     |
| {c}     | speed of light    | {#c}                    | None     |
| {e}     | elem. charge      | {#e}                    | None     |
| {eps_0} | permit. of vac.   | {#eps_0}                | None     |
| {G}     | grav. const.      | {#G}                    | None     |
| {g}     | grav. accel.      | {#g}                    | None     |
| {h}     | Planck const.     | {#h}                    | None     |
| {hbar}  | Reduced Pl. con.  | {#hbar}                 | None     |
| {H_0}   | Hubble const.     | {#H_0}                  | None     |
| {k_B}   | Boltzmann const.  | {#k_B}                  | None     |
| {k_e}   | Coulomb const.    | {#k_e}                  | None     |
| {L_sol} | Solar luminosity  | {#L_sol}                | None     |
| {M_sol} | Solar mass        | {#M_sol}                | None     |
| {mu_0}  | permeab. of vac.  | {#mu_0}                 | None     |
| {mu_B}  | Bohr magneton     | {#mu_B}                 | None     |
| {m_e}   | electron mass     | {#m_e}                  | None     |
| {m_p}   | proton mass       | {#m_p}                  | None     |
| {m_n}   | neutron mass      | {#m_n}                  | None     |
| {R_inf} | Rydberg constant  | {#R_inf}                | None     |
| {R_sol} | Solar radius      | {#R_sol}                | None     |
| {sigma} | Stef.Bolt. const. | {#sigma}                | None     |
| {N_A}   | Avogadro's const. | {#N_A}                  | None     |

## 8.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning                | Quantity scale          | Conversion factor |
|---------|------------------------|-------------------------|-------------------|
| <l>     | length                 | m                       | —                 |
| <m>     | mass                   | g                       | —                 |
| <t>     | time                   | s                       | —                 |
| <T>     | temperature            | K                       | —                 |
| <I>     | current                | A                       | —                 |
| <Iv>    | luminous intensity     | cd                      | —                 |
| <n>     | amount of substance    | mol                     | —                 |
| <the>   | angular displacement   | rad                     | —                 |
| <v>     | velocity               | m/s                     | —                 |
| <a>     | acceleration           | m/s <sup>2</sup>        | —                 |
| <jerk>  | jerk                   | m/s <sup>3</sup>        | —                 |
| <snap>  | snap                   | m/s <sup>4</sup>        | —                 |
| <crac>  | crackle                | m/s <sup>5</sup>        | —                 |
| <pop>   | pop                    | m/s <sup>6</sup>        | —                 |
| <rho>   | density                | kg/m <sup>3</sup>       | —                 |
| <rho_A> | area density           | kg/m <sup>2</sup>       | —                 |
| <rho_l> | linear density         | kg/m                    | —                 |
| <rho_E> | energy density         | J/m <sup>3</sup>        | —                 |
| <A>     | area                   | m <sup>2</sup>          | —                 |
| <V>     | volume                 | m <sup>3</sup>          | —                 |
| <p_mom> | momentum               | kg*m/s                  | —                 |
| <I_mom> | moment of inertia      | kg*m <sup>2</sup>       | —                 |
| <ome>   | angular velocity       | rad/s                   | —                 |
| <a_ome> | angular acceleration   | rad/s <sup>2</sup>      | —                 |
| <L_ome> | angular momentum       | kg*m <sup>2</sup> /s    | —                 |
| <F>     | force                  | N                       | —                 |
| <freq>  | frequency              | Hz                      | —                 |
| <J_i>   | impulse                | N*s                     | —                 |
| <Abs>   | absorbance             | m*s                     | —                 |
| <mu_v>  | dynamic viscosity      | Pa*s                    | —                 |
| <nu_v>  | kinematic viscosity    | m <sup>2</sup> /s       | —                 |
| <eps_s> | strain                 | 1                       | —                 |
| <sig_s> | stress                 | Pa                      | —                 |
| <E_Y>   | Young's modulus        | Pa                      | —                 |
| <k_s>   | spring constant        | N/m                     | —                 |
| <gam>   | surface tension        | J/m                     | —                 |
| <tau>   | torque                 | N*m                     | —                 |
| <E>     | energy                 | J                       | —                 |
| <S>     | entropy                | J/K                     | —                 |
| <p>     | pressure               | Pa                      | —                 |
| <mu_c>  | chemical potential     | J/mol                   | —                 |
| <W>     | work                   | J                       | —                 |
| <P>     | power                  | W                       | —                 |
| <E_sp>  | specific energy        | J/kg                    | —                 |
| <C_sp>  | specific heat capacity | J/(K*kg)                | —                 |
| <V_sp>  | specific volume        | m <sup>3</sup> /kg      | —                 |
| <Q>     | heat                   | J                       | —                 |
| <C_p>   | heat capacity          | J/K                     | —                 |
| <phi_Q> | heat flux density      | W/m <sup>2</sup>        | —                 |
| <C_mol> | molar concentration    | mol/m <sup>3</sup>      | —                 |
| <E_mol> | molar energy           | J/mol                   | —                 |
| <c_mol> | molar heat capacity    | J/(K*mol)               | —                 |
| <S_mol> | molar entropy          | J/(K*mol)               | —                 |
| <rr>    | reaction rate          | mol/(m <sup>3</sup> *s) | —                 |
| <kap>   | thermal conductance    | W/K                     | —                 |
| <lam_t> | thermal conductivity   | W/(m*K)                 | —                 |
| <R_t>   | thermal resistance     | K/W                     | —                 |
| <R_lam> | thermal resistivity    | K*m/W                   | —                 |
| <Q_vol> | volumetric flow rate   | m <sup>3</sup> /s       | —                 |

| Symbol    | Meaning                     | Quantity scale                                  | Conversion factor |
|-----------|-----------------------------|-------------------------------------------------|-------------------|
| <q>       | electric charge             | C                                               | —                 |
| <rho_q>   | electric charge density     | C/m <sup>3</sup>                                | —                 |
| <p_e>     | electric dipole moment      | C*m                                             | —                 |
| <p_qe>    | electric quadrupole moment  | C*m <sup>2</sup>                                | —                 |
| <D_e>     | electric displacement field | C/m <sup>2</sup>                                | —                 |
| <E_e>     | electric field strength     | V/m                                             | —                 |
| <V_e>     | electric field gradient     | V/m <sup>2</sup>                                | —                 |
| <alp_e>   | electric polarizability     | C <sup>2</sup> *m <sup>2</sup> *J <sup>-1</sup> | —                 |
| <alp_he>  | 1st hyperpolarizability     | C <sup>3</sup> *m <sup>3</sup> *J <sup>-2</sup> | —                 |
| <alp_hhe> | 2nd hyperpolarizability     | C <sup>4</sup> *m <sup>4</sup> *J <sup>-3</sup> | —                 |
| <G_e>     | electrical conductance      | S                                               | —                 |
| <sig_e>   | electrical conductivity     | S/m                                             | —                 |
| <phi_e>   | electric potential          | V                                               | —                 |
| <Phi_E>   | electric flux               | V*m                                             | —                 |
| <R>       | electrical resistance       | Ohm                                             | —                 |
| <rho_e>   | electrical resistivity      | Ohm*m                                           | —                 |
| <Mag>     | magnetization               | A/m                                             | —                 |
| <H>       | magnetic field strenght     | A/m                                             | —                 |
| <Phi_M>   | magnetic flux               | Wb                                              | —                 |
| <B>       | magnetic flux density       | T                                               | —                 |
| <mm>      | magnetic dipole moment      | N*m/T                                           | —                 |
| <mu>      | permeability                | H/m                                             | —                 |
| <eps>     | permittivity                | F/m                                             | —                 |
| <C>       | capacitance                 | F                                               | —                 |
| <J_c>     | current density             | A/m <sup>2</sup>                                | —                 |
| <Z>       | impedance                   | Ohm                                             | —                 |
| <L>       | inductance                  | H                                               | —                 |
| <Rel>     | reluctance                  | 1/H                                             | —                 |
| <P_o>     | optical power               | dpt                                             | —                 |
| <lam_w>   | wave length                 | m                                               | —                 |
| <k_n>     | wave number                 | 1/m                                             | —                 |
| <k_v>     | wave vector                 | 1/m                                             | —                 |
| <E_il>    | illuminance                 | lx                                              | —                 |
| <E_ir>    | irradiance                  | W/m <sup>2</sup>                                | —                 |
| <F_l>     | luinous flux                | lm                                              | —                 |
| <In>      | intensity                   | W/m <sup>2</sup>                                | —                 |
| <L_r>     | radiance                    | W/(m <sup>2</sup> *sr)                          | —                 |
| <I_r>     | radiant intensity           | W/sr                                            | —                 |
| <D_a>     | absorbed dose rate          | Gy/s                                            | —                 |
| <H_de>    | dose equivalent             | Sv                                              | —                 |
| <D_r>     | radiation dose              | Gy                                              | —                 |
| <A_r>     | radioactivity               | Bq                                              | —                 |
| <S_a>     | action                      | J/Hz                                            | —                 |
| <spin>    | spin                        | kg*m <sup>2</sup> *s <sup>-1</sup>              | —                 |

## 9 ESU: Electrostatic CGS units

Select this system with the leading ESU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

### 9.1 Derived units and aliases

| Symbol | Meaning | Definition or reference | Prefixes |
|--------|---------|-------------------------|----------|
| Gal    | Gal     | cm/s <sup>2</sup>       | None     |
| dyn    | dyne    | g*cm/s <sup>2</sup>     | All      |
| Ba     | Barye   | dyn/cm <sup>2</sup>     | None     |

| Symbol  | Meaning             | Definition or reference | Prefixes |
|---------|---------------------|-------------------------|----------|
| b       | Barye               | Ba                      | None     |
| erg     | erg                 | dyn*cm                  | None     |
| St      | Stokes              | cm <sup>2</sup> /s      | c        |
| P       | Poise               | g/(cm*s)                | c        |
| Ka      | Kayser              | cm <sup>-1</sup>        | None     |
| Pma     | permeability        | 1e-4*m <sup>2</sup>     | None     |
| Rayl    | Rayl                | g/(cm <sup>2</sup> *s)  | None     |
| radd    | radiation dose      | 100*erg/g               | None     |
| Fr      | franklin            | dyn <sup>1/2</sup> *cm  | None     |
| statC   | Statcoulomb         | Fr                      | None     |
| esu     | elstat. u.o. charge | Fr                      | None     |
| statA   | Statampere          | Fr/s                    | None     |
| statV   | Statvolt            | erg/Fr                  | None     |
| statOhm | Statohm             | s/cm                    | None     |
| statF   | Statfarad           | cm                      | None     |
| statH   | Stathenry           | s <sup>2</sup> /cm      | None     |
| statT   | Stattesla           | cm-3/2*g <sup>1/2</sup> | None     |
| statWb  | Statweber           | cm <sup>2</sup> *g      | None     |

## 9.2 System-specific constants

| Symbol  | Meaning           | Definition or reference                       | Prefixes |
|---------|-------------------|-----------------------------------------------|----------|
| {m_u}   | atom. mass const. | 1.660539069e-24*g                             | None     |
| {a_0}   | Bohr radius       | 5.291772105e-9*cm                             | None     |
| {k}     | Boltzmann const.  | 1.380649e-16*erg/K                            | None     |
| {m_e}   | electron mass     | 9.10938371e-28*g                              | None     |
| {alpha} | fine-str. const.  | 0.007297352564                                | None     |
| {G}     | gravit. const.    | 6.6743e-8*dyn*cm <sup>2</sup> /g <sup>2</sup> | None     |
| {h}     | Planck const.     | 6.62607015e-27*erg*s                          | None     |
| {hbar}  | reduced Pl. con.  | 1.054571817e-27*erg*s                         | None     |
| {c}     | speed of light    | 2.99792458e10*cm/s                            | None     |
| {pi}    | pi num.           | 3.1415926                                     | None     |
| {e}     | elementary charge | 4.80320471e-10*Fr                             | None     |
| {mu_B}  | Bohr magneton     | 2.780278273e-10*statA*cm <sup>2</sup>         | None     |

## 9.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol | Meaning              | Quantity scale | Conversion factor |
|--------|----------------------|----------------|-------------------|
| <l>    | length               | cm             | —                 |
| <m>    | mass                 | g              | —                 |
| <t>    | time                 | s              | —                 |
| <Iv>   | luminous intensity   | cd             | —                 |
| <n>    | amount of substance  | mol            | —                 |
| <the>  | angular displacement | rad            | —                 |
| <v>    | velocity             | cm/s           | —                 |
| <a>    | acceleration         | Gal            | —                 |
| <F>    | force                | dyn            | —                 |
| <E>    | energy               | erg            | —                 |
| <P>    | power                | erg/s          | —                 |
| <p>    | pressure             | Ba             | —                 |
| <mu_v> | dynamic viscosity    | P              | —                 |

| Symbol  | Meaning                     | Quantity scale | Conversion factor                      |
|---------|-----------------------------|----------------|----------------------------------------|
| <nu_v>  | kinematic viscosity         | St             | —                                      |
| <k_n>   | wave number                 | Ka             | —                                      |
| <q>     | electric charge             | statC          | $1\text{e}1/(\{c\}*100)$               |
| <I>     | current                     | statA          | $1\text{e}1/(\{c\}*100)$               |
| <phi_e> | electric potential          | statV          | $1\text{e}-8*(\{c\}*100)$              |
| <E_e>   | electric field strength     | statV/cm       | $1\text{e}-6*(\{c\}*100)$              |
| <D_e>   | electric displacement field | statC/cm2      | $1\text{e}5/(4*\{pi\}*(\{c\}*100))$    |
| <p_e>   | electric dipole moment      | statC*cm       | $1\text{e}-1/(\{c\}*100)$              |
| <Phi_E> | electric flux               | statC          | $1\text{e}1/(4*\{pi\}*(\{c\}*100))$    |
| <eps>   | permittivity                | 1              | $1\text{e}11/(4*\{pi\}*(\{c\}*100)^2)$ |
| <R>     | electrical resistance       | statOhm        | $(\{c\}*100)^2/1\text{e}9$             |
| <rho_e> | electrical resistivity      | statOhm*cm     | $(\{c\}*100)^2/1\text{e}11$            |
| <C>     | capacitance                 | statF          | $1\text{e}9/(\{c\}*100)^2$             |
| <L>     | inductance                  | statH          | $(\{c\}*100)^2/1\text{e}9$             |
| <B>     | magnetic flux density       | statT          | $1\text{e}-4*(\{c\}*100)$              |
| <H>     | magnetic field strenght     | statA/cm       | $1\text{e}3/(4*\{pi\}*(\{c\}*100))$    |
| <mm>    | magnetic dipole moment      | statA*cm2      | $1\text{e}-3/(\{c\}*100)$              |
| <Phi_M> | magnetic flux               | statWb         | $(\{c\}*100)*1\text{e}-8$              |
| <mu>    | permeability                | s2/cm2         | $4*\{pi\}*(\{c\}*100)^2*1\text{e}-7$   |

## 10 EMU: Electromagnetic CGS units

Select this system with the leading `EMU_` prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

### 10.1 Derived units and aliases

| Symbol | Meaning        | Definition or reference | Prefixes |
|--------|----------------|-------------------------|----------|
| Gal    | Gal            | cm/s2                   | None     |
| dyn    | dyne           | g*cm/s2                 | All      |
| Ba     | Barye          | dyn/cm2                 | None     |
| b      | Barye          | Ba                      | None     |
| erg    | erg            | dyn*cm                  | None     |
| St     | Stokes         | cm2/s                   | c        |
| P      | Poise          | g/(cm*s)                | c        |
| Ka     | Kayser         | cm-1                    | None     |
| Pma    | permeability   | 1e-4*m2                 | None     |
| Rayl   | Rayl           | g/(cm2*s)               | None     |
| radd   | radiation dose | 100*erg/g               | None     |
| G      | Gauss          | cm-1:2*g1:2*s-1         | None     |
| Oe     | Oersted        | g1:2/(cm1:2*s)          | None     |
| Gb     | Gilbert        | Oe*cm                   | None     |
| Mx     | Maxwell        | dyn1:2*cm2              | None     |
| Bi     | biot           | dyn1:2                  | None     |
| abA    | Abampere       | Bi                      | None     |
| abC    | Abcoulomb      | Bi*s                    | None     |
| aC     | Abcoulomb      | abC                     | None     |
| abV    | Abvolt         | g1:2*cm3:2*s-2          | None     |
| abOhm  | Abohm          | cm*s                    | None     |
| abF    | Abfarad        | cm-1*s2                 | None     |
| abH    | Abhenry        | cm                      | None     |

### 10.2 System-specific constants

| Symbol  | Meaning           | Definition or reference | Prefixes |
|---------|-------------------|-------------------------|----------|
| {m_u}   | atom. mass const. | 1.660539069e-24*g       | None     |
| {a_0}   | Bohr radius       | 5.291772105e-9*cm       | None     |
| {k}     | Boltzmann const.  | 1.380649e-16*erg/K      | None     |
| {m_e}   | electron mass     | 9.10938371e-28*g        | None     |
| {alpha} | fine-str. const.  | 0.007297352564          | None     |
| {G}     | gravit. const.    | 6.6743e-8*dyn*cm2/g2    | None     |
| {h}     | Planck const.     | 6.62607015e-27*erg*s    | None     |
| {hbar}  | reduced Pl. con.  | 1.054571817e-27*erg*s   | None     |
| {c}     | speed of light    | 2.99792458e10*cm/s      | None     |
| {pi}    | pi num.           | 3.1415926               | None     |
| {mu_B}  | Bohr magneton     | 9.274010066e-21*erg/G   | None     |
| {e}     | elementary charge | 1.602176634e-20*abC     | None     |

### 10.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning                     | Quantity scale | Conversion factor |
|---------|-----------------------------|----------------|-------------------|
| <l>     | length                      | cm             | —                 |
| <m>     | mass                        | g              | —                 |
| <t>     | time                        | s              | —                 |
| <Iv>    | luminous intensity          | cd             | —                 |
| <n>     | amount of substance         | mol            | —                 |
| <the>   | angular displacement        | rad            | —                 |
| <v>     | velocity                    | cm/s           | —                 |
| <a>     | acceleration                | Gal            | —                 |
| <F>     | force                       | dyn            | —                 |
| <E>     | energy                      | erg            | —                 |
| <P>     | power                       | erg/s          | —                 |
| <p>     | pressure                    | Ba             | —                 |
| <mu_v>  | dynamic viscosity           | P              | —                 |
| <nu_v>  | kinematic viscosity         | St             | —                 |
| <k_n>   | wave number                 | Ka             | —                 |
| <B>     | magnetic flux density       | G              | 1e-4              |
| <H>     | magnetic field strenght     | Oe             | 1e3/(4*{pi})      |
| <mm>    | magnetic dipole moment      | erg/G          | 1e-3              |
| <Phi_M> | magnetic flux               | Mx             | 1e-8              |
| <mu>    | permeability                | 1              | 4*{pi}*1e-7       |
| <q>     | electric charge             | abC            | 1e1               |
| <I>     | current                     | abA            | 1e1               |
| <phi_e> | electric potential          | abV            | 1e-8              |
| <E_e>   | electric field strength     | abV/cm         | 1e-6              |
| <D_e>   | electric displacement field | abC/cm2        | 1e5/(4*{pi})      |
| <p_e>   | electric dipole moment      | abC*cm         | 1e-1              |
| <Phi_E> | electric flux               | abC            | 1e1/(4*{pi})      |
| <eps>   | permittivity                | s2/cm2         | 1e11/(4*{pi})     |
| <R>     | electrical resistance       | abOhm          | 1e-9              |
| <rho_e> | electrical resistivity      | abOhm*cm       | 1e-11             |
| <C>     | capacitance                 | abF            | 1e9               |
| <L>     | inductance                  | abH            | 1e-9              |

## 11 GU: Gaussian CGS units

Select this system with the leading GU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared

table legend.

## 11.1 Derived units and aliases

| Symbol  | Meaning             | Definition or reference                               | Prefixes |
|---------|---------------------|-------------------------------------------------------|----------|
| Gal     | Gal                 | cm/s <sup>2</sup>                                     | None     |
| dyn     | dyne                | g*cm/s <sup>2</sup>                                   | All      |
| Ba      | Barye               | dyn/cm <sup>2</sup>                                   | None     |
| b       | Barye               | Ba                                                    | None     |
| erg     | erg                 | dyn*cm                                                | None     |
| St      | Stokes              | cm <sup>2</sup> /s                                    | c        |
| P       | Poise               | g/(cm*s)                                              | c        |
| Ka      | Kayser              | cm <sup>-1</sup>                                      | None     |
| Pma     | permeability        | 1e-4*m <sup>2</sup>                                   | None     |
| Rayl    | Rayl                | g/(cm <sup>2</sup> *s)                                | None     |
| radd    | radiation dose      | 100*erg/g                                             | None     |
| Fr      | franklin            | dyn <sup>1:2</sup> *cm                                | None     |
| statC   | Statcoulomb         | Fr                                                    | None     |
| esu     | elstat. u.o. charge | Fr                                                    | None     |
| statA   | Statampere          | Fr/s                                                  | None     |
| statV   | Statvolt            | erg/Fr                                                | None     |
| statOhm | Statohm             | s/cm                                                  | None     |
| statF   | Statfarad           | cm                                                    | None     |
| statH   | Stathenry           | s <sup>2</sup> /cm                                    | None     |
| G       | Gauss               | cm <sup>-1:2</sup> *g <sup>1:2</sup> *s <sup>-1</sup> | None     |
| Oe      | Oersted             | g <sup>1:2</sup> /(cm <sup>1:2</sup> *s)              | None     |
| Gb      | Gilbert             | Oe*cm                                                 | None     |
| Mx      | Maxwell             | dyn <sup>1:2</sup> *cm <sup>2</sup>                   | None     |

## 11.2 System-specific constants

| Symbol  | Meaning           | Definition or reference                       | Prefixes |
|---------|-------------------|-----------------------------------------------|----------|
| {m_u}   | atom. mass const. | 1.660539069e-24*g                             | None     |
| {a_0}   | Bohr radius       | 5.291772105e-9*cm                             | None     |
| {k}     | Boltzmann const.  | 1.380649e-16*erg/K                            | None     |
| {m_e}   | electron mass     | 9.10938371e-28*g                              | None     |
| {alpha} | fine-str. const.  | 0.007297352564                                | None     |
| {G}     | gravit. const.    | 6.6743e-8*dyn*cm <sup>2</sup> /g <sup>2</sup> | None     |
| {h}     | Planck const.     | 6.62607015e-27*erg*s                          | None     |
| {hbar}  | reduced Pl. con.  | 1.054571817e-27*erg*s                         | None     |
| {c}     | speed of light    | 2.99792458e10*cm/s                            | None     |
| {pi}    | pi num.           | 3.1415926                                     | None     |
| {e}     | elementary charge | 4.80320471e-10*Fr                             | None     |
| {mu_B}  | Bohr magneton     | 9.274010066e-21*erg/G                         | None     |

## 11.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol | Meaning            | Quantity scale | Conversion factor |
|--------|--------------------|----------------|-------------------|
| <l>    | length             | cm             | —                 |
| <m>    | mass               | g              | —                 |
| <t>    | time               | s              | —                 |
| <Iv>   | luminous intensity | cd             | —                 |

| Symbol  | Meaning                     | Quantity scale | Conversion factor               |
|---------|-----------------------------|----------------|---------------------------------|
| <n>     | amount of substance         | mol            | —                               |
| <the>   | angular displacement        | rad            | —                               |
| <v>     | velocity                    | cm/s           | —                               |
| <a>     | acceleration                | Gal            | —                               |
| <F>     | force                       | dyn            | —                               |
| <E>     | energy                      | erg            | —                               |
| <P>     | power                       | erg/s          | —                               |
| <p>     | pressure                    | Ba             | —                               |
| <mu_v>  | dynamic viscosity           | P              | —                               |
| <nu_v>  | kinematic viscosity         | St             | —                               |
| <k_n>   | wave number                 | Ka             | —                               |
| <q>     | electric charge             | statC          | $1e1/(\{c\}*100)$               |
| <I>     | current                     | statA          | $1e1/(\{c\}*100)$               |
| <phi_e> | electric potential          | statV          | $1e-8*(\{c\}*100)$              |
| <E_e>   | electric field strength     | statV/cm       | $1e-6*(\{c\}*100)$              |
| <D_e>   | electric displacement field | statC/cm2      | $1e5/(4*\{pi\}*(\{c\}*100))$    |
| <p_e>   | electric dipole moment      | statC*cm       | $1e-1/(\{c\}*100)$              |
| <Phi_E> | electric flux               | statC          | $1e1/(4*\{pi\}*(\{c\}*100))$    |
| <eps>   | permittivity                | 1              | $1e11/(4*\{pi\}*(\{c\}*100)^2)$ |
| <R>     | electrical resistance       | statOhm        | $(\{c\}*100)^2/1e9$             |
| <rho_e> | electrical resistivity      | statOhm*cm     | $(\{c\}*100)^2/1e11$            |
| <C>     | capacitance                 | statF          | $1e9/(\{c\}*100)^2$             |
| <L>     | inductance                  | statH          | $(\{c\}*100)^2/1e9$             |
| <B>     | magnetic flux density       | G              | $1e-4$                          |
| <H>     | magnetic field strenght     | Oe             | $1e3/(4*\{pi\})$                |
| <mm>    | magnetic dipole moment      | erg/G          | $1e-3$                          |
| <Phi_M> | magnetic flux               | Mx             | $1e-8$                          |
| <mu>    | permeability                | 1              | $4*\{pi\}*1e-7$                 |

## 12 IU: Imperial units

Select this system with the leading IU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

### 12.1 Derived units and aliases

| Symbol | Meaning        | Definition or reference | Prefixes |
|--------|----------------|-------------------------|----------|
| twip   | twip           | $17.6388888e-6*m$       | None     |
| th     | thou           | $25.4e-6*m$             | None     |
| mil    | mil            | th                      | None     |
| in     | inch           | $0.0254*m$              | None     |
| ''     | inch           | in                      | None     |
| li     | link           | $0.201168*m$            | None     |
| ft     | foot           | $0.3048*m$              | None     |
| '      | foot           | ft                      | None     |
| yd     | yard           | $0.9144*m$              | None     |
| rd     | rod/pole/perch | $5.0292*m$              | None     |
| ch     | chain          | $20.1168*m$             | None     |
| fur    | furlong        | $201.168*m$             | None     |
| mi     | mile           | $1609.344*m$            | None     |
| lea    | league         | $4828.032*m$            | None     |
| le     | league         | lea                     | None     |
| ftm    | fathom         | $1.8288*m$              | None     |
| cb     | cable          | $219.456*m$             | None     |
| nmi    | nautical mile  | $1852*m$                | None     |
| NM     | nautical mile  | nmi                     | None     |

| Symbol | Meaning            | Definition or reference                         | Prefixes |
|--------|--------------------|-------------------------------------------------|----------|
| ac     | acre               | 4046.8564224*m2                                 | None     |
| gr     | grain              | 64.79891e-3*g                                   | None     |
| dr     | drachm/dram        | 1.771845195*g                                   | None     |
| oz     | ounce              | 28.349523125*g                                  | None     |
| lb     | pound              | 453.59237*g                                     | None     |
| cwt    | long hundredweight | 50.80234544e3*g                                 | None     |
| ton    | long ton           | 1.0160469088e6*g                                | None     |
| slug   | slug               | 14.59390294e3*g                                 | None     |
| min    | minute             | 60*s                                            | None     |
| h      | hour               | 60*min                                          | None     |
| day    | day                | 24*h                                            | None     |
| yr_t   | tropical year      | 365.24219*day                                   | k, m, G  |
| yr_j   | Julian year        | 365.25*day                                      | k, m, G  |
| yr_g   | Gregorian year     | 365.2425*day                                    | k, m, G  |
| yr     | year               | yr_j                                            | k, m, G  |
| degR   | degree Rankine     | 5/9*K                                           | None     |
| mph    | miles per hour     | mi/h                                            | None     |
| lbf    | pound-force        | 4.448222e3*g*m*s-2                              | None     |
| Cel    | degree Celsius     | K (temperature reference;<br>offset conversion) | None     |
| degF   | degree Fahrenheit  | K (temperature reference;<br>offset conversion) | None     |
| bc     | barleycorn         | 8.4667e-3*m                                     | None     |
| hh     | hand               | 0.1016*m                                        | None     |
| sqrd   | sq. rod/po./per.   | 25.29285264*m2                                  | None     |
| ro     | rood               | 1011.7141056*m2                                 | None     |
| sqmi   | square mile        | 2589988.110336*m2                               | None     |
| minim  | minim              | 59.1938802083e-9*m3                             | None     |
| fls    | fluid scruple      | 1.18387760416e-6*m3                             | None     |
| fldr   | fluid drachm       | 3.5516328125e-6*m3                              | None     |
| floz   | fluid ounce        | 28.4130625e-6*m3                                | None     |
| gi     | gill               | 142.0653125e-6*m3                               | None     |
| pt     | pint               | 568.26125e-6*m3                                 | None     |
| qt     | quart              | 1136.5225e-6*m3                                 | None     |
| gal    | gallon             | 4.54609e-3*m3                                   | None     |
| bblo   | oil barrel         | 0.16365924*m3                                   | None     |
| st     | stone              | 6.35029318e3*g                                  | None     |
| qr     | quarter            | 12.70058636e3*g                                 | None     |

## 12.2 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol | Meaning              | Quantity scale | Conversion factor |
|--------|----------------------|----------------|-------------------|
| <l>    | length               | ft             | —                 |
| <m>    | mass                 | lb             | —                 |
| <t>    | time                 | s              | —                 |
| <Iv>   | luminous intensity   | cd             | —                 |
| <n>    | amount of substance  | mol            | —                 |
| <the>  | angular displacement | rad            | —                 |
| <V>    | volume               | gal            | —                 |
| <A>    | area                 | ac             | —                 |
| <v>    | velocity             | mph            | —                 |

## 13 US: United States customary units

Select this system with the leading `US_` prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

### 13.1 Derived units and aliases

| Symbol  | Meaning            | Definition or reference                         | Prefixes |
|---------|--------------------|-------------------------------------------------|----------|
| twip    | twip               | 17.6388888e-6*m                                 | None     |
| th      | thou               | 25.4e-6*m                                       | None     |
| mil     | mil                | th                                              | None     |
| in      | inch               | 0.0254*m                                        | None     |
| ''      | inch               | in                                              | None     |
| li      | link               | 0.201168*m                                      | None     |
| ft      | foot               | 0.3048*m                                        | None     |
| '       | foot               | ft                                              | None     |
| yd      | yard               | 0.9144*m                                        | None     |
| rd      | rod/pole/perch     | 5.0292*m                                        | None     |
| ch      | chain              | 20.1168*m                                       | None     |
| fur     | furlong            | 201.168*m                                       | None     |
| mi      | mile               | 1609.344*m                                      | None     |
| lea     | league             | 4828.032*m                                      | None     |
| le      | league             | lea                                             | None     |
| ftm     | fathom             | 1.8288*m                                        | None     |
| cb      | cable              | 219.456*m                                       | None     |
| nmi     | nautical mile      | 1852*m                                          | None     |
| NM      | nautical mile      | nmi                                             | None     |
| ac      | acre               | 4046.8564224*m2                                 | None     |
| gr      | grain              | 64.79891e-3*g                                   | None     |
| dr      | drachm/dram        | 1.771845195*g                                   | None     |
| oz      | ounce              | 28.349523125*g                                  | None     |
| lb      | pound              | 453.59237*g                                     | None     |
| cwt     | long hundredweight | 50.80234544e3*g                                 | None     |
| ton     | long ton           | 1.0160469088e6*g                                | None     |
| slug    | slug               | 14.59390294e3*g                                 | None     |
| min     | minute             | 60*s                                            | None     |
| h       | hour               | 60*min                                          | None     |
| day     | day                | 24*h                                            | None     |
| yr_t    | tropical year      | 365.24219*day                                   | k, m, G  |
| yr_j    | Julian year        | 365.25*day                                      | k, m, G  |
| yr_g    | Gregorian year     | 365.2425*day                                    | k, m, G  |
| yr      | year               | yr_j                                            | k, m, G  |
| degR    | degree Rankine     | 5/9*K                                           | None     |
| mph     | miles per hour     | mi/h                                            | None     |
| lbf     | pound-force        | 4.448222e3*g*m*s-2                              | None     |
| Cel     | degree Celsius     | K (temperature reference;<br>offset conversion) | None     |
| degF    | degree Fahrenheit  | K (temperature reference;<br>offset conversion) | None     |
| p       | point              | 352.778e-6*m                                    | None     |
| P       | pica               | 4.233e-3*m                                      | None     |
| sqft    | sq. survey foot    | 0.09290341*m2                                   | None     |
| sqch    | square chain       | 404.68564224*m2                                 | None     |
| section | section            | 2.589998e6*m2                                   | None     |
| twp     | survey township    | 93.23993e6*m2                                   | None     |
| cuin    | cubic inch         | 16.387064e-6*m3                                 | None     |
| cuft    | cubic foot         | 28.316846592e-3*m3                              | None     |
| cuyd    | cubic yard         | 0.764554857984*m3                               | None     |
| acft    | acre-foot          | 1233.482*m3                                     | None     |

| Symbol | Meaning          | Definition or reference | Prefixes |
|--------|------------------|-------------------------|----------|
| minim  | minim            | 61.611519921875e-9*m3   | None     |
| fldr   | fluid drachm     | 3.6966911953125e-6*m3   | None     |
| tsp    | teaspoon         | 4.92892159375e-6*m3     | None     |
| tbsp   | tablespoon       | 14.78676478125e-6*m3    | None     |
| floz   | fluid ounce      | 29.5735295625e-6*m3     | None     |
| jig    | jig              | 44.36029434375e-6*m3    | None     |
| gi     | gill             | 118.29411825e-6*m3      | None     |
| c      | cup              | 236.5882365e-6*m3       | None     |
| pt     | pint             | 0.473176473e-3*m3       | None     |
| qt     | quart            | 0.946352946e-3*m3       | None     |
| pot    | pottle           | 1.892705892e-3*m3       | None     |
| gal    | gallon           | 3.785411784e-3*m3       | None     |
| bbl    | liquid barrel    | 119.240471196e-3*m3     | None     |
| bblo   | oil barrel       | 158.987294928e-3*m3     | None     |
| hhd    | hogshead         | 238.480942392e-3*m3     | None     |
| ptd    | dry pint         | 0.55061047135749e-3*m3  | None     |
| qtd    | dry quart        | 1.1012209427149e-3*m3   | None     |
| gald   | dry gallon       | 4.4048837708599e-3*m3   | None     |
| pk     | peck             | 8.8097675417199e-3*m3   | None     |
| bu     | bushel           | 35.239070166879e-3*m3   | None     |
| bbll   | liquid barrel    | 115.62712358400e-3*m3   | None     |
| cwts   | short hundredwe. | 45.359237e3*g           | None     |
| tons   | short ton        | 907.18474e3*g           | None     |
| dwt    | pennyweight      | 1.55517384*g            | None     |
| ozt    | troy ounce       | 31.1034768*g            | None     |
| lbt    | troy pound       | 373.241721*g            | None     |

## 13.2 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol | Meaning              | Quantity scale | Conversion factor |
|--------|----------------------|----------------|-------------------|
| <l>    | length               | ft             | —                 |
| <m>    | mass                 | lb             | —                 |
| <t>    | time                 | s              | —                 |
| <Iv>   | luminous intensity   | cd             | —                 |
| <n>    | amount of substance  | mol            | —                 |
| <the>  | angular displacement | rad            | —                 |
| <V>    | volume               | gal            | —                 |
| <A>    | area                 | ac             | —                 |
| <v>    | velocity             | mph            | —                 |

## 14 AU: Atomic units

Select this system with the leading AU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

### 14.1 Derived units and aliases

| Symbol | Meaning | Definition or reference           | Prefixes |
|--------|---------|-----------------------------------|----------|
| E_h    | hartree | $\{\hbar\}^2/(\{m_e\}*\{a_0\}^2)$ | None     |

## 14.2 System-specific constants

| Symbol         | Meaning           | Definition or reference                      | Prefixes |
|----------------|-------------------|----------------------------------------------|----------|
| $\{\hbar\}$    | reduced Pl. con.  | 1                                            | None     |
| $\{e\}$        | elementary charge | 1                                            | None     |
| $\{m_e\}$      | electr. rest mass | 1                                            | None     |
| $\{\kappa_0\}$ | permittivity      | 1                                            | None     |
| $\{a_0\}$      | Bohr radius       | $\{\kappa_0\}*\{\hbar\}^2/(\{m_e\}*\{e\}^2)$ | None     |
| $\{\alpha\}$   | fine-str. const.  | 1/137                                        | None     |
| $\{c\}$        | speed of light    | $\{\alpha\}-1*\{a_0\}*E_h/\{\hbar\}$         | None     |
| $\{r_e\}$      | class. e. radius  | $\{\alpha\}^2*\{a_0\}$                       | None     |
| $\{\lambda\}$  | red. Com. wavele. | $\{\alpha\}*\{a_0\}$                         | None     |
| $\{m_p\}$      | proton mass       | $1836*\{m_e\}$                               | None     |

## 14.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol    | Meaning                    | Quantity scale                | Conversion factor |
|-----------|----------------------------|-------------------------------|-------------------|
| <l>       | length                     | $\{a_0\}$                     | —                 |
| <m>       | mass                       | $\{m_e\}$                     | —                 |
| <t>       | time                       | $\{\hbar\}/E_h$               | —                 |
| <I>       | current                    | $\{e\}*E_h/\{\hbar\}$         | —                 |
| <Iv>      | luminous intensity         | cd                            | —                 |
| <n>       | amount of substance        | mol                           | —                 |
| <the>     | angular displacement       | rad                           | —                 |
| <v>       | velocity                   | $\{a_0\}*E_h/\{\hbar\}$       | —                 |
| <p_mom>   | momentum                   | $\{\hbar\}/\{a_0\}$           | —                 |
| <rho_q>   | electric charge density    | $\{e\}/\{a_0\}^3$             | —                 |
| <q>       | electric charge            | $\{e\}$                       | —                 |
| <p_e>     | electric dipole moment     | $\{e\}*\{a_0\}$               | —                 |
| <p_qe>    | electric quadrupole moment | $\{e\}*\{a_0\}^2$             | —                 |
| <phi_e>   | electric potential         | $E_h/\{e\}$                   | —                 |
| <E_e>     | electric field strength    | $E_h/\{e\}*\{a_0\}$           | —                 |
| <V_e>     | electric field gradient    | $E_h/\{e\}*\{a_0\}^2$         | —                 |
| <eps>     | permittivity               | $\{\kappa_0\}$                | —                 |
| <alp_e>   | electric polarizability    | $\{e\}^2*\{a_0\}^2/E_h$       | —                 |
| <alp_he>  | 1st hyperpolarizability    | $\{e\}^3*\{a_0\}^3/E_h^2$     | —                 |
| <alp_hhe> | 2nd hyperpolarizability    | $\{e\}^4*\{a_0\}^4/E_h^3$     | —                 |
| <B>       | magnetic flux density      | $\{\hbar\}/(\{e\}*\{a_0\}^2)$ | —                 |
| <mm>      | magnetic dipole moment     | $\{\hbar\}*\{e\}/\{m_e\}$     | —                 |
| <E>       | energy                     | $E_h$                         | —                 |
| <F>       | force                      | $E_h/\{a_0\}$                 | —                 |
| <S_a>     | action                     | $\{\hbar\}$                   | —                 |

## 15 PU: Planck units

Select this system with the leading PU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

## 15.1 Derived units and aliases

This system defines its characteristic scales through the quantity symbols below rather than additional named unit symbols.

## 15.2 System-specific constants

| Symbol | Meaning          | Definition or reference | Prefixes |
|--------|------------------|-------------------------|----------|
| {c}    | speed of light   | 1                       | None     |
| {G}    | grav. const.     | 1                       | None     |
| {hbar} | reduced Pl. con. | 1                       | None     |
| {k_B}  | Boltzmann const. | 1                       | None     |

## 15.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning      | Quantity scale                                                                 | Conversion factor |
|---------|--------------|--------------------------------------------------------------------------------|-------------------|
| <l>     | length       | $(\{\text{hbar}\} * \{\text{G}\} / \{\text{c}\}^3)^{1:2}$                      | —                 |
| <m>     | mass         | $(\{\text{hbar}\} * \{\text{c}\} / \{\text{G}\})^{1:2}$                        | —                 |
| <t>     | time         | $(\{\text{hbar}\} * \{\text{G}\} / \{\text{c}\}^5)^{1:2}$                      | —                 |
| <T>     | temperature  | $(\{\text{hbar}\} * \{\text{c}\}^5 * \{\text{k}_B\} - 2 / \{\text{G}\})^{1:2}$ | —                 |
| <A>     | area         | $\{\text{hbar}\} * \{\text{G}\} / \{\text{c}\}^3$                              | —                 |
| <V>     | volume       | $(\{\text{hbar}\} * \{\text{G}\} * \{\text{c}\}^{-3})^{3:2}$                   | —                 |
| <p_mom> | momentum     | $(\{\text{hbar}\} * \{\text{c}\}^3 / \{\text{G}\})^{1:2}$                      | —                 |
| <E>     | energy       | $(\{\text{hbar}\} * \{\text{c}\}^5 / \{\text{G}\})^{1:2}$                      | —                 |
| <F>     | force        | $\{\text{c}\}^4 / \{\text{G}\}$                                                | —                 |
| <rho>   | density      | $\{\text{c}\}^5 / \{\text{hbar}\} * \{\text{G}\}^{-2}$                         | —                 |
| <a>     | acceleration | $(\{\text{c}\}^7 / \{\text{hbar}\} / \{\text{G}\})^{1:2}$                      | —                 |

## 16 SRU: Units for special relativity

Select this system with the leading SRU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

## 16.1 Derived units and aliases

This system defines its characteristic scales through the quantity symbols below rather than additional named unit symbols.

## 16.2 System-specific constants

| Symbol | Meaning        | Definition or reference | Prefixes |
|--------|----------------|-------------------------|----------|
| {c}    | speed of light | 1                       | None     |

## 16.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning          | Quantity scale | Conversion factor |
|---------|------------------|----------------|-------------------|
| <l>     | length           | m              | —                 |
| <m>     | mass             | kg             | —                 |
| <t>     | time             | m              | {#c}-1            |
| <E>     | energy           | kg             | {#c}2             |
| <p_mom> | momentum         | kg             | {#c}              |
| <v>     | velocity         | 1              | {#c}              |
| <L_ome> | angular momentum | kg*m           | {#c}              |
| <F>     | force            | kg/m           | {#c}2             |
| <a>     | acceleration     | m-1            | {#c}2             |
| <rho_E> | energy density   | kg/m3          | {#c}2             |

## 17 GRU: Units for general relativity

Select this system with the leading `GRU_` prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

### 17.1 Derived units and aliases

| Symbol | Meaning      | Definition or reference                                 | Prefixes |
|--------|--------------|---------------------------------------------------------|----------|
| eV     | electronvolt | $1.602176634e-19 \text{ m}^2 \text{ kg} \text{ s}^{-2}$ | M, G, T  |

### 17.2 System-specific constants

| Symbol  | Meaning          | Definition or reference               | Prefixes      |
|---------|------------------|---------------------------------------|---------------|
| {c}     | speed of light   | 1                                     | None          |
| {hbar}  | reduced Pl. con. | 1                                     | None          |
| {eps_0} | permit. of vac.  | 1                                     | None          |
| {k_B}   | Boltzmann const. | 1                                     | None          |
| {pi}    | pi num.          | {#pi}                                 | None          |
| {alpha} | fine-str. const. | $\frac{e^2}{4\pi \epsilon_0 \hbar c}$ | {#hbar}*{#c}) |
| {e}     | elem. charge     | $(4\pi \epsilon_0 \hbar c \alpha)$    | 1:2 None      |

### 17.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning          | Quantity scale | Conversion factor |
|---------|------------------|----------------|-------------------|
| <l>     | length           | GeV-1          | {#hbar}*{#c}      |
| <m>     | mass             | GeV            | {#c}-2            |
| <t>     | time             | GeV-1          | {#hbar}           |
| <E>     | energy           | GeV            | —                 |
| <p_mom> | momentum         | GeV            | {#c}-1            |
| <v>     | velocity         | 1              | {#c}              |
| <L_ome> | angular momentum | 1              | {#hbar}           |
| <A>     | area             | GeV-2          | ({#hbar}*{#c})2   |
| <F>     | force            | GeV2           | ({#hbar}*{#c})-1  |
| <rho_E> | energy density   | GeV4           | ({#hbar}*{#c})-3  |
| <q>     | electric charge  | {#e}/{#e}      | —                 |

## 18 GEO: Geometrized units

Select this system with the leading `GEO_` prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

### 18.1 Derived units and aliases

This system defines its characteristic scales through the quantity symbols below rather than additional named unit symbols.

### 18.2 System-specific constants

| Symbol | Meaning        | Definition or reference | Prefixes |
|--------|----------------|-------------------------|----------|
| {c}    | speed of light | 1                       | None     |
| {G}    | grav. const.   | 1                       | None     |

### 18.3 Quantity symbols and conversion factors

`<name>` denotes the quantity scale in the third column. Where provided, the last column defines `|name|`, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning          | Quantity scale | Conversion factor |
|---------|------------------|----------------|-------------------|
| <l>     | length           | m              | —                 |
| <m>     | mass             | m              | {#c}2/{#G}        |
| <t>     | time             | m              | {#c}-1            |
| <E>     | energy           | m              | {#c}4/{#G}        |
| <p_mom> | momentum         | m              | {#c}3/{#G}        |
| <v>     | velocity         | 1              | {#c}              |
| <L_ome> | angular momentum | m2             | {#c}3/{#G}        |
| <F>     | force            | 1              | {#c}4/{#G}        |
| <a>     | acceleration     | m-1            | {#c}2             |
| <rho_E> | energy density   | m-2            | {#c}4/{#G}        |